

● EASY

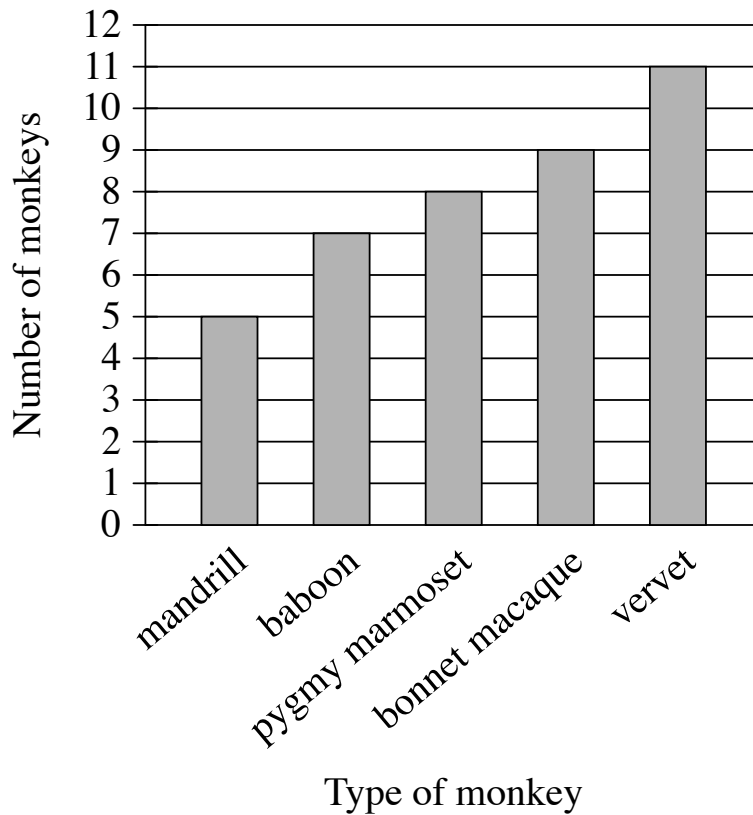
● PROBLEM-SOLVING AND DATA ANALYSIS

One-variable data: Distributions and measures of center and spread

02

10 questions · ~15 min

The bar graph shows the number of each type of monkey in a sanctuary.



- The Number of monkeys data for the 5 bars are as follows:
 - mandrill: 5
 - baboon: 7
 - pygmy marmoset: 8
 - bonnet macaque: 9
 - vervet: 11

How many more vervets are in this sanctuary than mandrills?

A. 11

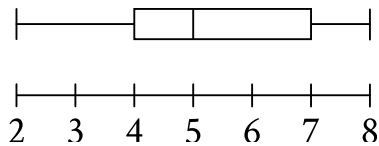
B. 6

C. 5

D. 3

Q2

PROBLEM-SOLVING AND DATA ANALYSIS



- From left to right the values of the vertical bars in the box plot are as follows:
 - First vertical bar: 2
 - Second vertical bar: 4
 - Third vertical bar: 5
 - Fourth vertical bar: 7
 - Fifth vertical bar: 8

The box plot summarizes 15 data values. What is the median of this data set?

A. 2

B. 3

C. 5

D. 8

Number of High School Students Who
Completed Summer Internships

High school	Year				
	2008	2009	2010	2011	2012
Foothill	87	80	75	76	70
Valley	44	54	65	76	82
Total	131	134	140	152	152

The table above shows the number of students from two different high schools who completed summer internships in each of five years. No student attended both schools. Which of the following statements are true about the number of students who completed summer internships for the 5 years shown?

1. The mean number from Foothill High School is greater than the mean number from Valley High School.
 2. The median number from Foothill High School is greater than the median number from Valley High School.
- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

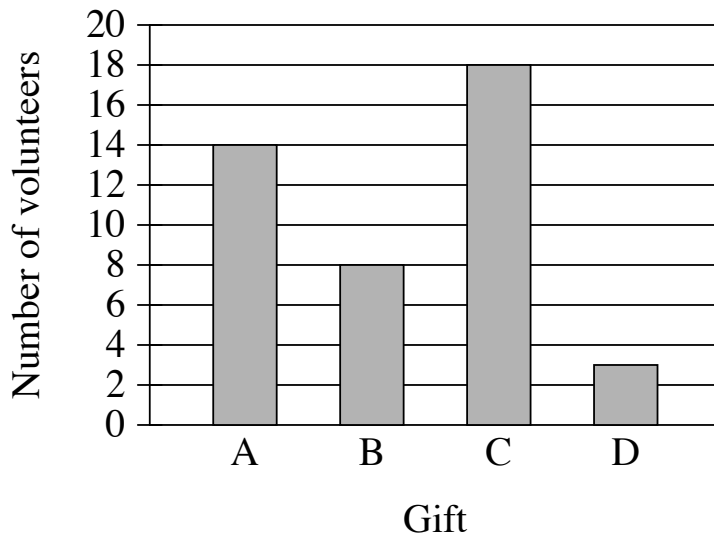
Q4

PROBLEM-SOLVING AND DATA ANALYSIS

71, 72, 73, 76, 77, 79, 83, 87, 93

What is the median of the data shown?

- A. 71
- B. 77
- C. 78
- D. 79



- The Number of volunteers data for the 4 bars are as follows:
 - A: 14
 - B: 8
 - C: 18
 - D: 3

In April, there were 43 volunteers in a cleanup project. Each volunteer was asked to choose a small gift labeled A, B, C, or D. The bar graph shows the number of volunteers who chose each gift. How many volunteers chose gift C?

- A. 3
- B. 8
- C. 14
- D. 18

Data set A: 1, 2, 3, 4, 5, 6, 7

Data set B: 1, 1, 2, 2, 3, 3, 4

Which of the following statements correctly compares the means of data set A and data set B?

- A. The mean of each data set is 2.
- B. The mean of each data set is 4.
- C. The mean of data set A is less than the mean of data set B.
- D. The mean of data set A is greater than the mean of data set B.

Q7

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

Each value in the data set shown represents the height, in centimeters, of a plant.

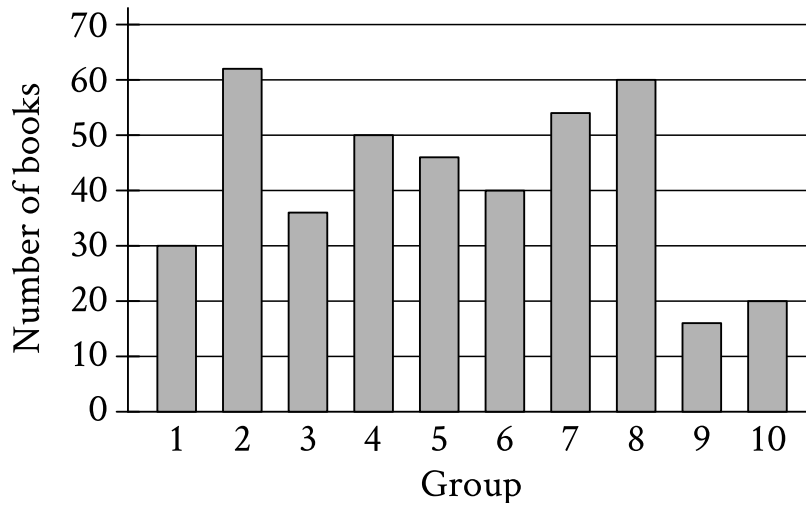
6, 10, 13, 2, 15, 22, 10, 4, 4, 4

What is the mean height, in centimeters, of these plants?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The data for the 10 categories are as follows:
 - Group 1: 30
 - Group 2: 62
 - Group 3: 36
 - Group 4: 50
 - Group 5: 46
 - Group 6: 40
 - Group 7: 54
 - Group 8: 60
 - Group 9: 16
 - Group 10: 20

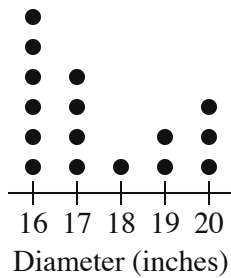
The bar graph shows the distribution of 414 books collected by 10 different groups for a book drive. How many books were collected by group 1?

STUDENT-PRODUCED RESPONSE

3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

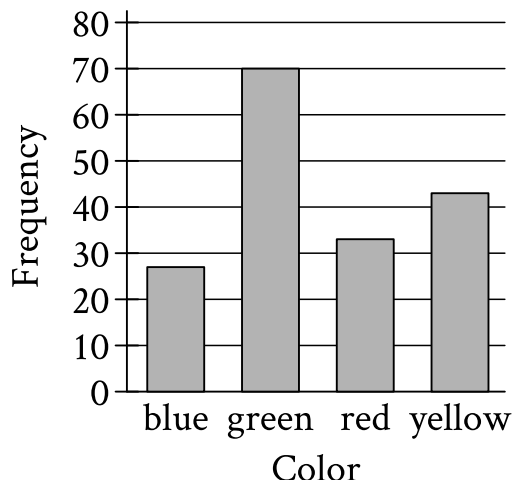
To study the characteristics of sea stars in a group of tide pools, researchers measured the diameter of the sea stars within the tide pools. The dot plot gives the diameter, to the nearest inch, of each of the sea stars in these tide pools.



- The data for the dot plot are as follows:
 - 16: 6 dots
 - 17: 4 dots
 - 18: 1 dot
 - 19: 2 dots
 - 20: 3 dots

Based on the dot plot, how many sea stars had a diameter, to the nearest inch, of 16 inches?

- A. 16
- B. 6
- C. 4
- D. 1



- The Frequency data for the 4 bars are as follows:
 - blue: 27
 - green: 70
 - red: 33
 - yellow: 43

A data set consists of 173 colors. The bar graph shows the number of times each color appears in the data set. Which color appears 70 times?

- A. Blue
- B. Green
- C. Red
- D. Yellow